

Krutik Tikam

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🐙 GitHub | 🔗 LinkedIn

PROFESSIONAL SUMMARY

Applied AI Engineer specializing in computer vision, continuous-control reinforcement learning, and signal processing. Proven ability to build and deploy real-time tracking pipelines, active learning subsystems, and predictive simulations using PyTorch, YOLO, and Gymnasium. Focused on leveraging modern developer tools like Claude and Cursor to optimize neural architectures for high-throughput headless execution and low-latency inference in production environments.

TECHNICAL SKILLS

Computer Vision & RL: PyTorch, YOLO, ByteTrack, OpenCV, Reinforcement Learning (PPO), Diffusion Models, Signal Processing

Data & Machine Learning: XGBoost, Scikit-Learn, MNE-Python, ChromaDB, Pandas, Active Learning

Systems & Simulation: Python, FastAPI, Gymnasium, Blender API, Docker, Cursor, Claude

PROFESSIONAL EXPERIENCE

Hudl India Private Limited

Mumbai, India

Wyscout Analyst

Aug 2026 – Present

- Analyze high-frequency sports telemetry and broadcast data to support advanced computer vision and predictive analytics workflows.
- Leverage proprietary rule frameworks to structure and tag kinematic spatio-temporal features, ensuring high-quality data ingestion for machine learning models.

Talent Corner HR Services Private Limited

Mumbai, India

Software Engineering Intern

Dec 2025 – Feb 2026

- Architected and optimized **PostgreSQL** database schemas to efficiently manage and query a dense relational dataset of **60,000+** professional profiles, significantly minimizing search latency.
- Engineered asynchronous REST APIs using **FastAPI** and Python to power a dynamic authentication system and an automated document generation engine.
- Leveraged AI coding assistants to accelerate backend pipeline development, deploying via Git and conducting rigorous API load testing to ensure high-availability system stability.

VISION, ROBOTICS & SIGNAL AI PROJECTS

Sports Analyser: CV Tracking & Active Learning | *YOLO, ByteTrack, OpenCV, ChromaDB*

- Architected a real-time computer vision pipeline tracking multi-agent kinematics across broadcast footage using YOLO object detection and ByteTrack Kalman filtering.
- Developed an active learning subsystem to automatically extract high-entropy video frames, iteratively pre-labeling them to continuously improve model confidence.
- Engineered a semantic vector RAG engine using ChromaDB to match real-time kinematic spatio-temporal features against 70+ proprietary sports rules.

Robotic Arm RL Platform: Headless Simulation | *PyTorch, PPO, Gymnasium, Blender API*

- Constructed a high-performance, headless 3D simulation environment achieving execution speeds exceeding 800 FPS for Proximal Policy Optimization (PPO) training.
- Engineered complex continuous reward functions incorporating precision grip multipliers and spatial penalties to optimize automated telemetry and model convergence.
- Designed automated auditing scripts using Cursor to rapidly verify rigid-body mechanical tolerances and actuator tracking accuracy for production-grade policy deployment.

NeuroRehab-BCI: Real-Time Signal Classification | *PyTorch, MNE-Python, FastAPI*

- Developed an end-to-end signal processing pipeline and custom temporal-spatial convolutional neural network (TS-CNN) to classify multi-channel EEG signals.
- Engineered a virtual hardware bridge to stream live data windows to a FastAPI inference server, achieving highly optimized real-time prediction latency.

Social-Media AI Image Generator: Low-VRAM Optimization | *PyTorch, Diffusers, Accelerate*

- Engineered a local diffusion-model generation pipeline optimized to run entirely on low-tier consumer hardware without triggering memory fault errors.
- Utilized native PyTorch Scaled Dot-Product Attention and CPU-offloading architecture to significantly constrain peak operational VRAM during high-fidelity image synthesis.

EDUCATION

Nagandas Khandwala College

Mumbai, Maharashtra

B.Sc. Computer Science (Spl. in Artificial Intelligence & Machine Learning)

2023 – 2026

- **Academic Standing:** CGPA: 7.5
- **Relevant Focus:** Data Structures & Algorithms (DSA), Neural Networks, Database Management Systems (DBMS), Distributed Web Architecture.

CERTIFICATIONS

Cloud Engineering: AWS Data Engineering for Generative AI Applications, AWS Connecting Systems and Machines for Industrial Manufacturing, Microsoft Azure Cloud Computing (SDP)

Machine Learning & Data Science: DeepLearning.AI Machine Learning Specialization, TensorFlow Developer Certificate